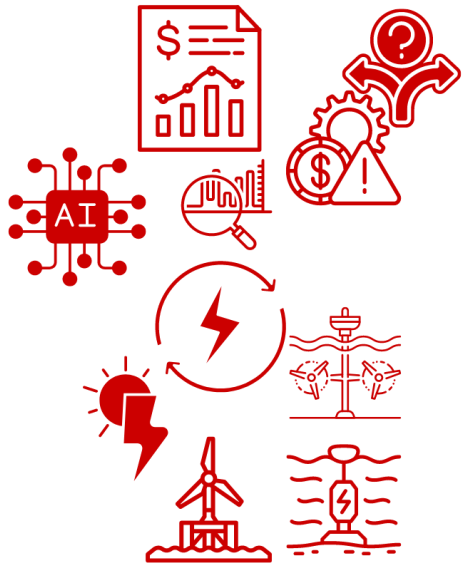
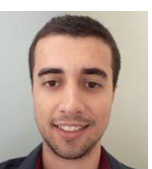




Exploring the Feasibility of Co-Located Marine Renewable Energy Systems: A Mooring System and Economic Analysis in the North Carolina Coast

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PhD Students:
Neda Jamaledin, Victor Faria



April, 2024

Offshore Wind Project Activity as of 5/31/23

Legend:

- Lease Areas:** Green
- Wind Energy Areas:** Orange
- Call Areas:** Yellow
- Dormant Wind Areas:** Grey
- Bathymetry Depth (meters):**
 - ≤ 30
 - 30-45
 - 45-60
 - 60-1300
 - ≥ 1300

Regions and Key Projects:

- Northeast:** New England Aqua Ventus 1, Maine Research Array, Gulf of Maine Call Area, Bay State Wind, Revolution Wind, Rhode Island, South Fork Wind Farm, Block Island Wind Farm, Sunrise Wind Farm, Sunrise Wind Farm (Residual), New England Wind 2, New England Wind 1, Vineyard Wind 1, Beacon Wind 1, SouthCoast Wind 2, SouthCoast Wind 1a, SouthCoast Wind 1b, SouthCoast Wind (Residual), Floating Demonstration, Vineyard Northeast.
- Mid Atlantic:** Empire Wind 1, Empire Wind 2, Atlantic Shores Offshore Wind North, Atlantic Shores Offshore Wind South (Project 1), Atlantic Shores Offshore Wind South (Project 2), Ocean Wind 1, Ocean Wind 2, Skipjack 1, Skipjack 2, Marlin, Monomah Wind, MarWin (Residual), Coastal Virginia Offshore Wind (Pilot), Coastal Virginia Offshore Wind (Commercial), Kitty Hawk North, Kitty Hawk South, Vineyard Mid-Atlantic, Blueprint Wind, Attentive Energy, Community Offshore Wind, Atlantic Shores Offshore Wind Right, Leading Light Wind, Central Atlantic Call Areas.
- Pacific:** Oahu North Call Area, Oahu South Call Area, Cocos Bay Call Area, Brookings Call Area, North Coast (Humboldt) CSP, North Coast (Humboldt) RWE, Total Energies, Duke Energy, Central Coast (Morro Bay) Inventory, Central Coast (Morro Bay) EDP/RENTECH, Central Coast (Morro Bay) Equator, CADEMO.
- Gulf of Mexico:** Galveston 1 WEA, Galveston 2 WEA, Lake Charles WEA.

- The US currently has around 17GW of offshore wind power offtake agreements.
- States Planning Targets and Mandates: 63GW by 2040 and 112GW by 2050
- In NC, Governor Cooper issued in 2021 the Executive Order No. 218 with targets for 2.8GW of offshore wind by 2030 and 8GW by 2040
- NC has other considerable offshore resources like wave and ocean currents. Can we take advantage of the future offshore wind energy infrastructure to integrate these other resources? This work looks at wave energy!

The infographic shows the following data for the United States:

Region	Distance (miles)
West Coast	250
Inland States	41
Gulf Coast	84
East Coast	460
Contiguous United States	830

Legend:

- Wave (Blue)
- Tidal (Orange)
- Ocean Current (Light Blue)
- Ocean Thermal (Grey)
- River (Dark Blue)

DOE (2023). Offshore Wind Market Report: 2023 Edition.
NREL (2021). Marine Energy in the United States: An Overview of Opportunities.

Portfolio Optimization

Find Optimal Location of Devices
(Cost x Energy to Shore)

Inputs:

- Historical Wave Height-Period (WAVEWATCH III)
- Historical Wind Speed Data (NREL)
- Wave Device Designs (RM3, Pelamis)
- Wind Device Designs (8, 12, 15, 18MW)
- BOEM Lease/Planning Areas

Model:

**Objective Function-
Energy Maximization to
Shore**

$$\max_{(y,v)} \sum_{t \in T} (EG_t^{Wind} + EG_t^{Wave} - \Delta_t) \quad (1)$$

**Budget Constraint: Wind +
Wave + Transmission**

$$\frac{TAC^{Wind} + TAC^{Wave} + TC}{8760 \sum_{t \in T} (EG_t^{Wind} + EG_t^{Wave} - \Delta_t)} \leq LCOE \quad (2)$$

Curtailment Constraint

$$EG_t^{Wind} + EG_t^{Wave} - \Delta_t \leq RP_{Tr} \quad \forall t \in T \quad (3)$$

**Maximum Number of
Devices Per Site Location**

$$y_i^{Wind} \leq \overline{NT}_i^{Wind} \quad \forall i \in I_{Wind}^d \quad (4)$$

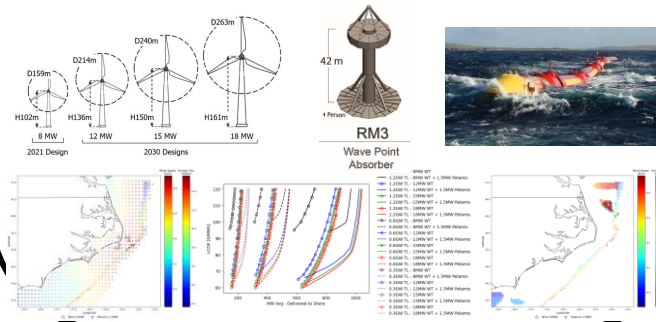
**Maximum Number of
Devices Per Site Location**

$$y_i^{Wave} \leq \overline{NT}_i^{Wave} \quad \forall i \in I_{Wave}^d \quad (5)$$

**Maximum Radius for
Energy Collection System**

$$\sum_{k \in OR_{I_i}^{Wind}} y_i^{Wind} + \sum_{k \in OR_{I_i}^{Wave}} y_i^{Wave} \leq (1 - v_i) \overline{NT} \quad \forall i \in I_{Tr} \quad (6)$$

$$\sum_{i \in I_{Tr}} v_i = 1 \quad (7)$$

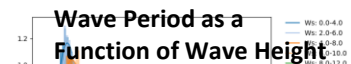


Wave Device Under Extreme Condition

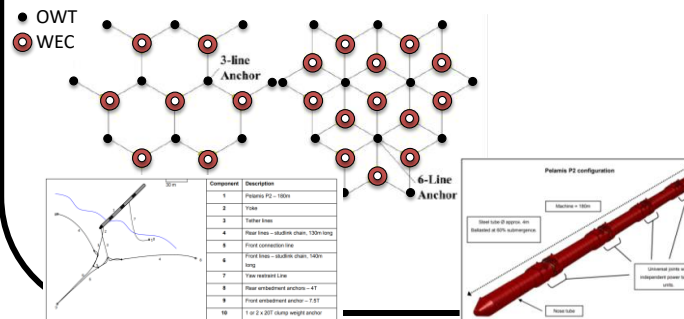
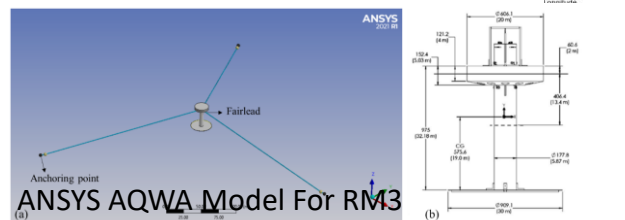
Evaluate How Wave Devices Get
Affected By Extreme Conditions

- 1) Extreme Value Analysis To Estimate Critical Ocean Conditions For Wave Design

$$f_{U_w, H_s, T_p}(u, h, t) \approx f_{U_w}(u) f_{H_s|U_w}(h|u) f_{T_p|H_s}(t|h)$$



- 2) Mooring System Design



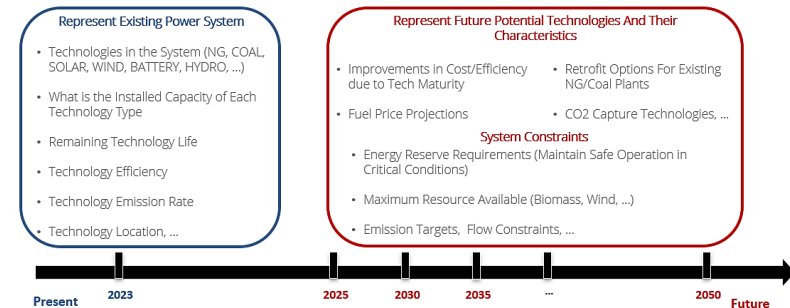
Capacity Expansion

How the NC Energy System May Evolve In The Future
2023-2050 (Deterministic and Stochastic Simulations)

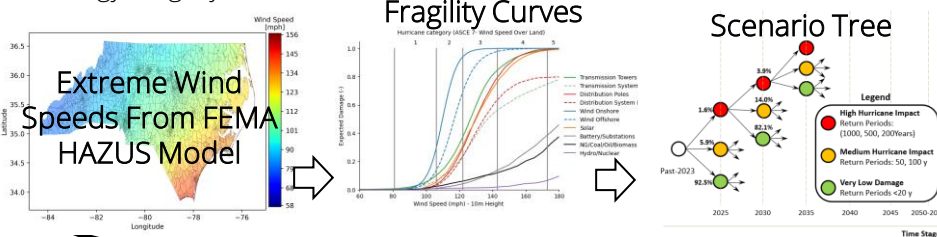
- **Deterministic Model:** Runs Without Risk of Hurricane Damage – How Technologies Compete With Each Other
- **Stochastic Model:** Run Considering Technology Vulnerability to Hurricanes – How the Risk of Hurricane May Affect The Solution

Inputs:

- **(Deterministic):** Existing Generators Capacity; Emissions; Efficiencies, Fuel Prices, Technology Prices, Improvements in CFs, Max Capacity, Max Activity, Energy Imports, Energy Reserve Requirements/Capacity Credits, Technology Life, Government Policies/CO2 Targets, etc.



- **(Stochastic) = (Deterministic) + Synthetic Hurricane Wind Fields, Technology Fragility Curves**



Portfolio Optimization

- For a given budget (**LCOE**), find me the solution that delivers the most electricity to the shore
- Solution: Number of Turbines at Each Viable Site Location (**y**), Location of Offshore Substation (**v**)
- Take into consideration:
 - Curtailment
 - Transmissions system cost
 - Wind Energy Cost
 - Wave Energy Cost
 - Maximum Radius for Collecting Energy

Simplified Model Formulation

**Objective Function-
Energy Maximization to
Shore**

$$\text{Max}_{(y,v)} \sum_{t \in T} (EG_t^{Wind} + EG_t^{Wave} - \Delta_t) \quad (1)$$

Budget Constraint

$$\frac{TAC^{Wind} + TAC^{Wave} + TC}{8760 \sum_{t \in T} (EG_t^{Wind} + EG_t^{Wave} - \Delta_t)} \leq \overline{LCOE} \quad (2)$$

$$TAC^{Wind} = \sum_{i \in I_{Wind}} AC_i^{Wind} y_i^{Wind} \quad (3)$$

Aggregating Costs

$$TAC^{Wave} = \sum_{i \in I_{Wave}} AC_i^{Wave} y_i^{Wave} \quad (4)$$

$$TC = \sum_{i \in I_{Tr}} v_i TC_i \quad (5)$$

**Aggregating Energy
Generation**

$$EG_t^{Wind} = \sum_{i \in I_{Wind}} EG_{i,t}^{Wind} y_i^{Wind} \quad \forall t \in T \quad (6)$$

$$EG_t^{Wave} = \sum_{i \in I_{Wave}} EG_{i,t}^{Wave} y_i^{Wave} \quad \forall t \in T \quad (7)$$

Curtailment Constraint

$$EG_t^{Wind} + EG_t^{Wave} - \Delta_t \leq RP_{Tr} \quad \forall t \in T \quad (8)$$

**Maximum Number of
Devices Per Site Location**

$$y_{i,d}^{Wind} \leq \overline{NT}_{i,d}^{Wind} \dots \quad \forall i \in I_{Wind} \quad (9)$$

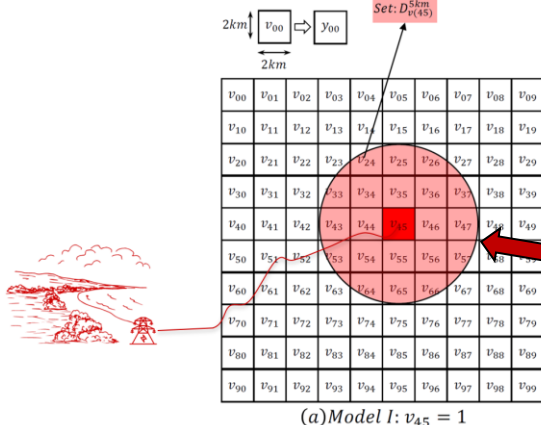
$$y_i^{Wave} \leq \overline{NT}_i^{Wave} \dots \quad \forall i \in I_{Wave} \quad (10)$$

**Maximum Radius for
Energy Collection System**

$$\sum_{k \in OR_{v_i}^{Wind}} y_i^{Wind} + \sum_{k \in OR_{v_i}^{Wave}} y_i^{Wave} \leq \quad \forall i \in I_{Tr} \quad (11)$$

$$(1 - v_i) \overline{NT} \quad (12)$$

$$\sum_{i \in I_{Tr}} v_i = 1$$



Variables

y_i^{Wind} : Number of wind turbines deployed at the site $i \in I_{Wind}$, $y_i^{Wind} \in \mathbb{Z}^+$

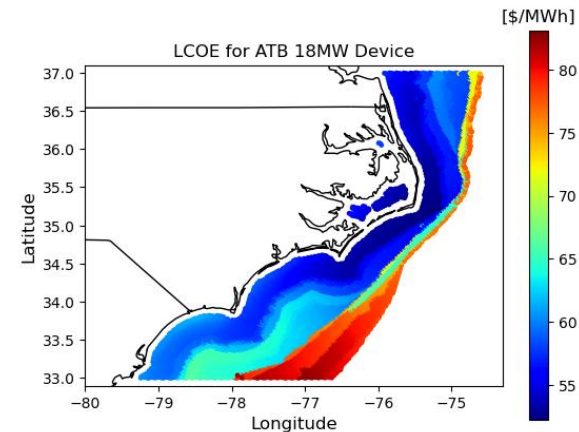
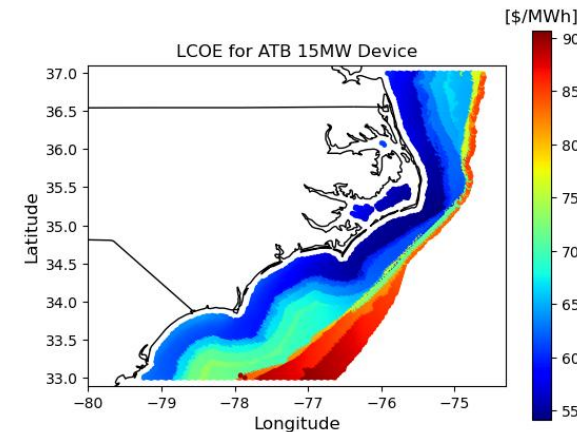
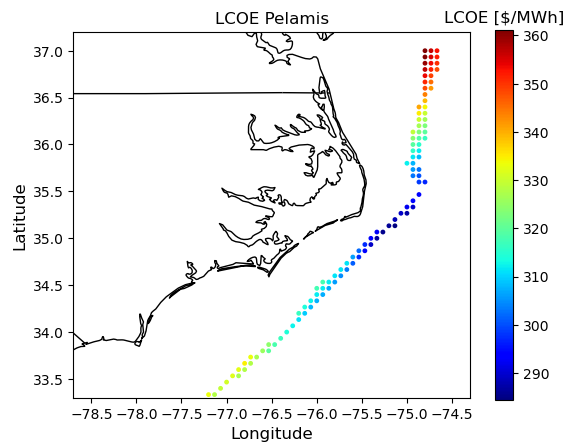
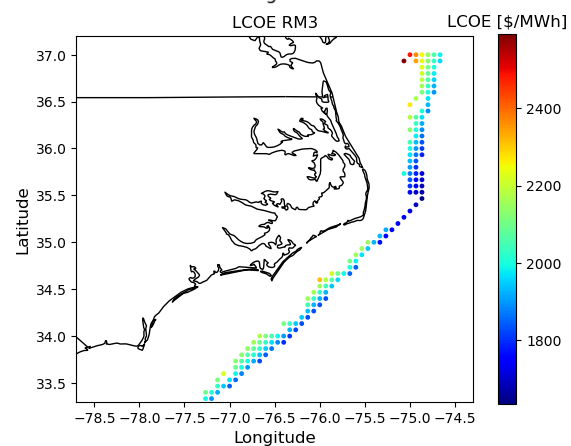
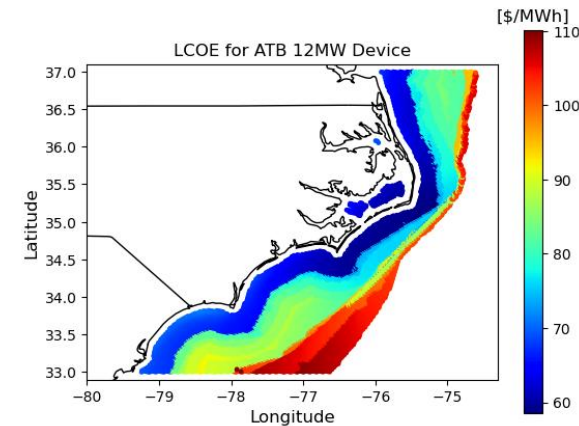
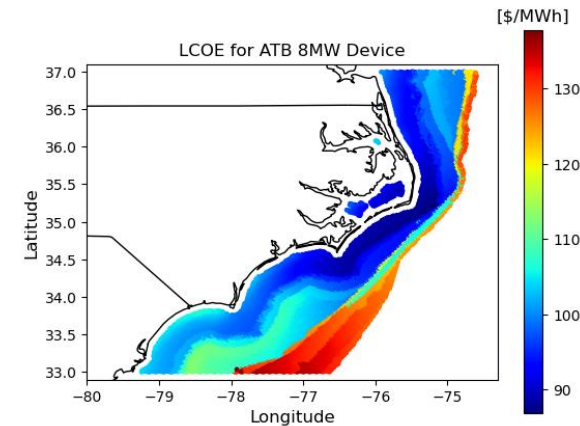
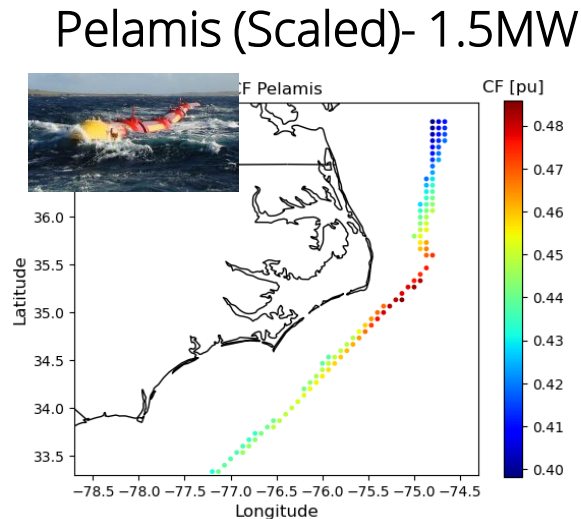
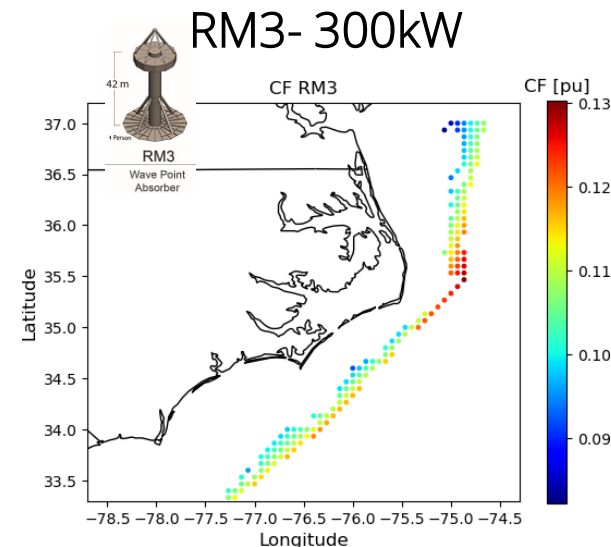
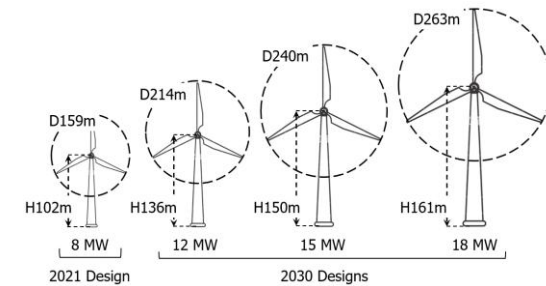
y_i^{Wave} : Number of wave turbines deployed at the site $i \in I_{Wave}$, $y_i^{Wave} \in \mathbb{Z}^+$

v_i : Binary variable responsible for controlling the center of the energy collection system for each energy technology, $v_i \in \{0,1\}$

Portfolio Optimization Inputs

For Wave Designs: RM3 not suitable for NC Region

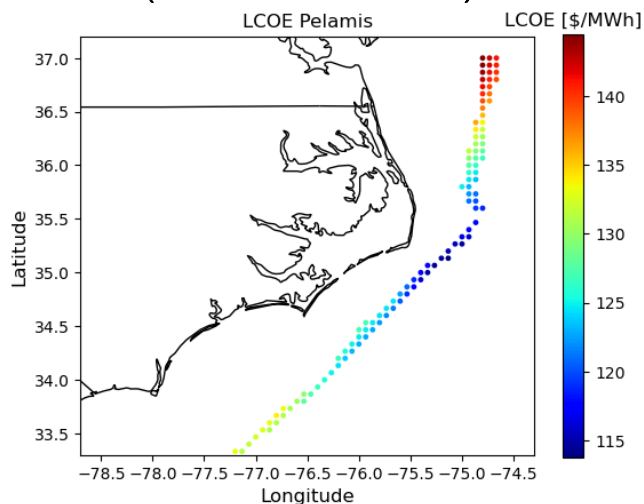
For Wind Designs: Costs Expected to Decrease Fast by 2030



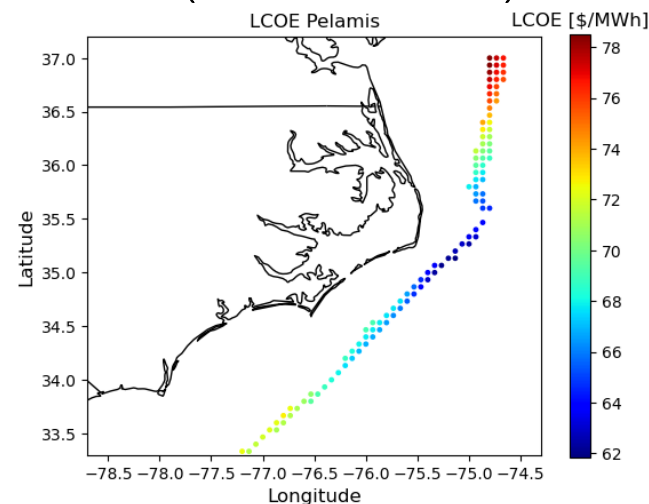
Portfolio Optimization Inputs

- Even with the current scaled Pelamis (290-360 \$/MWh) it is hard to compete with offshore wind (140-55 \$/MWh)!
- Looking at a recent expert elicitation study by NREL (Baca et al., 2022):
 - Estimate **2.5x** reduction in WEC cost by 2050 as a **conservative** estimate
 - Estimate **4.6x** reduction in WEC cost by 2050 as an **optimistic** estimate

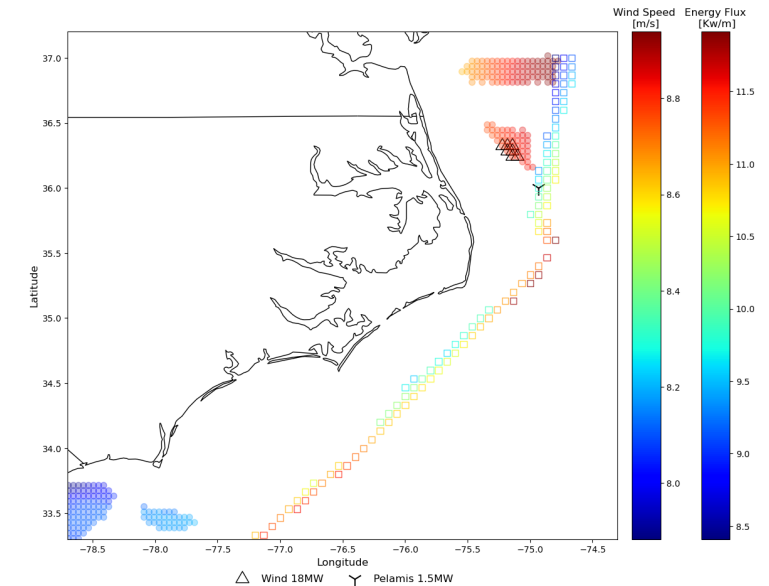
Conservative-2050 Pelamis
(2.5x reduction in cost)



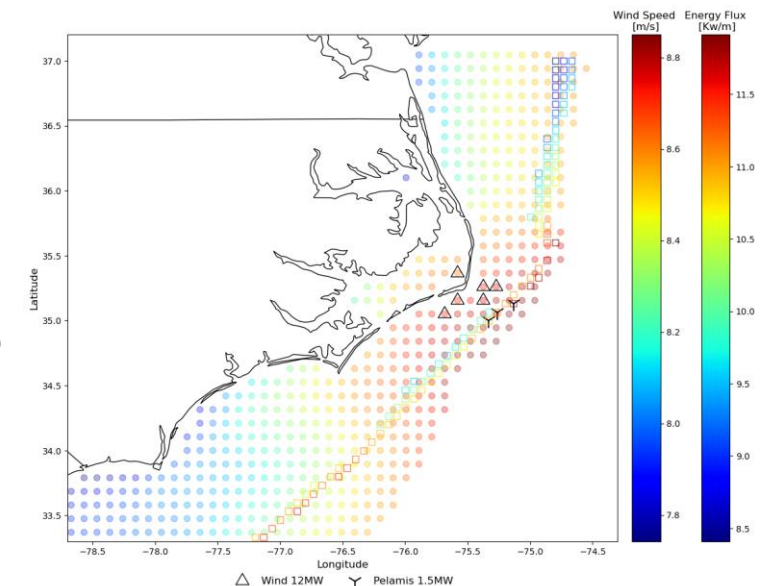
Optimistic-2050 Pelamis
(4.6x reduction in cost)



Deploy Wind
from BOEM
Areas



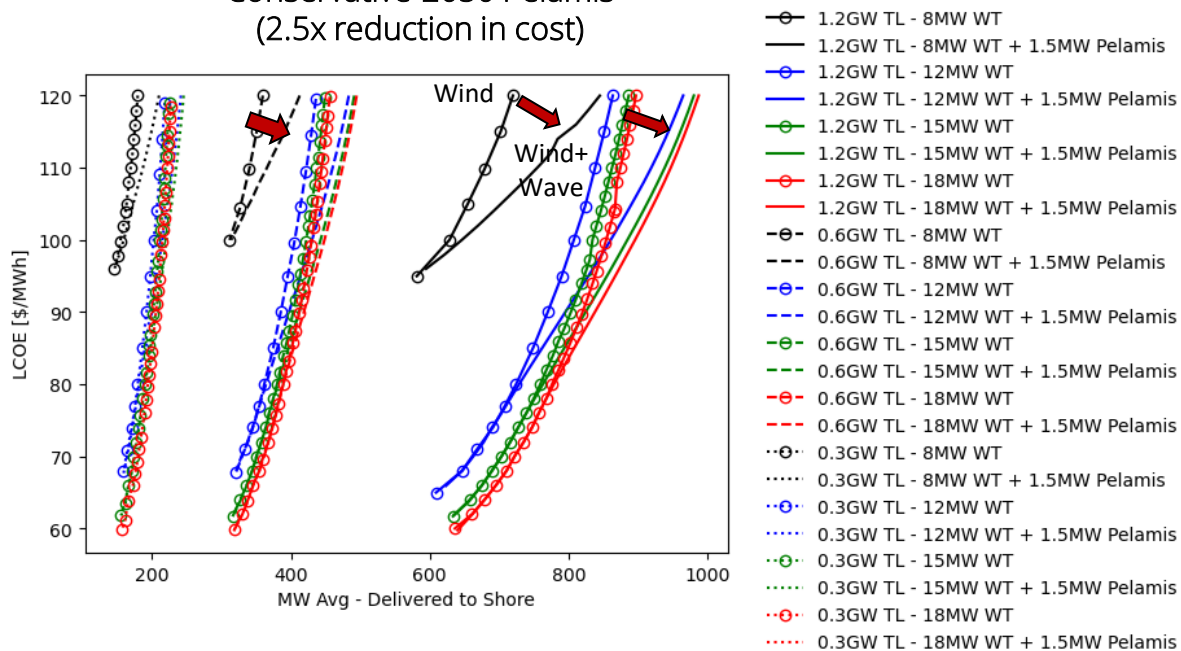
Deploy Wind
from Any Site
<1000m Deep



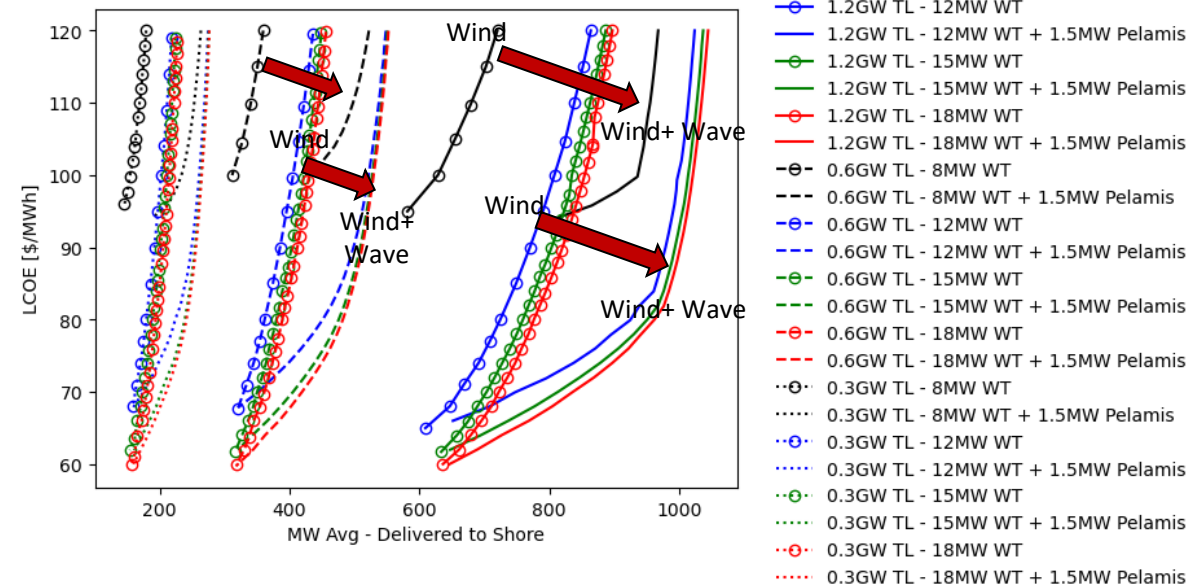
Portfolio Optimization Results (NOT BOEM LIMITED)

- Simulating cases with different wind turbine designs (8, 12, 15, 18MW) and transmission line capacities (1.2, 0.6, 0.3GW)
- Assuming the conservative and optimistic cost reduction from NREL we see that the wave energy devices get used by the model as an alternative to deliver more energy to the shore making a **better use of the existing transmission system infrastructure**.

Conservative-2050 Pelamis
(2.5x reduction in cost)

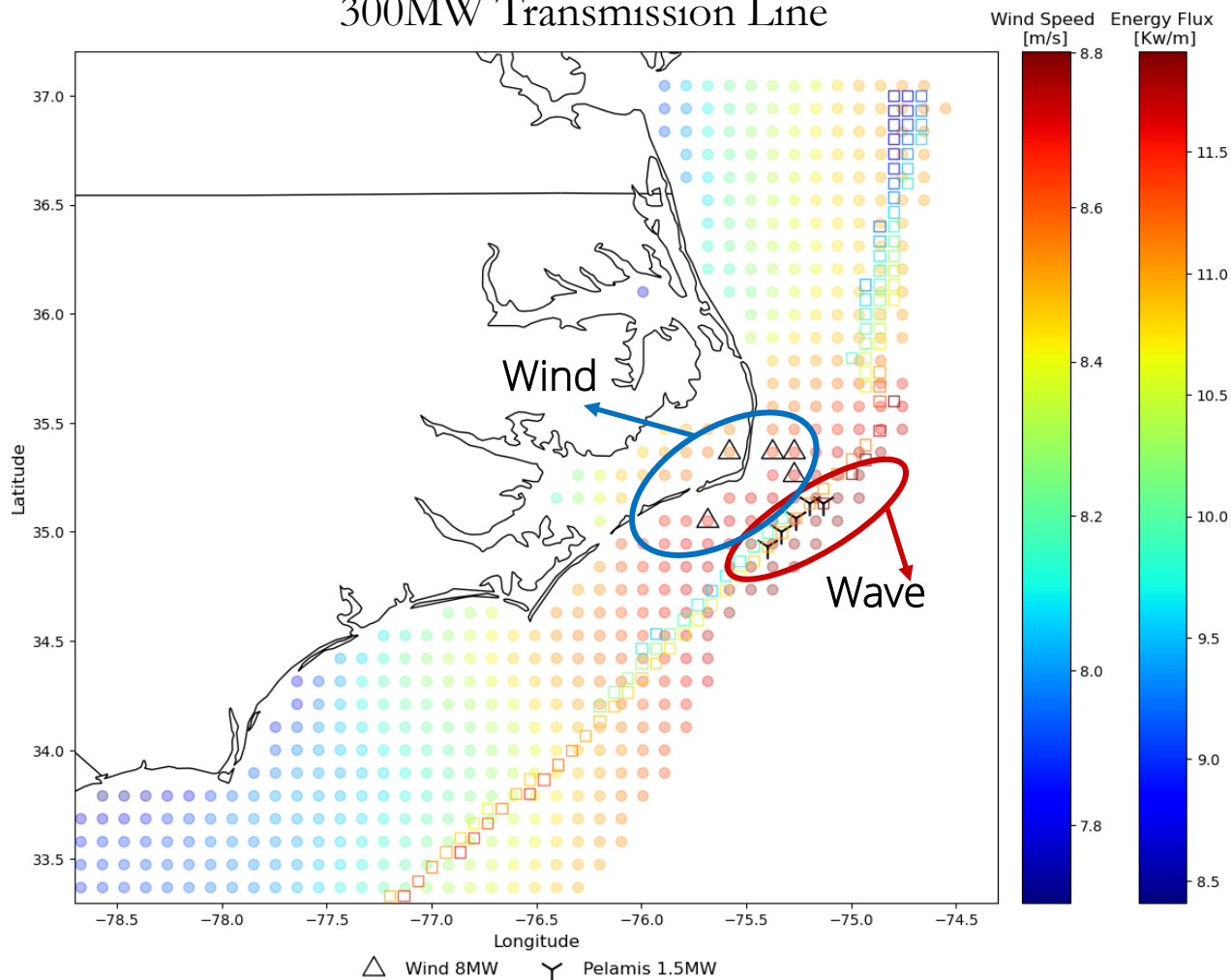


Optimistic-2050 Pelamis
(4.6x reduction in cost)

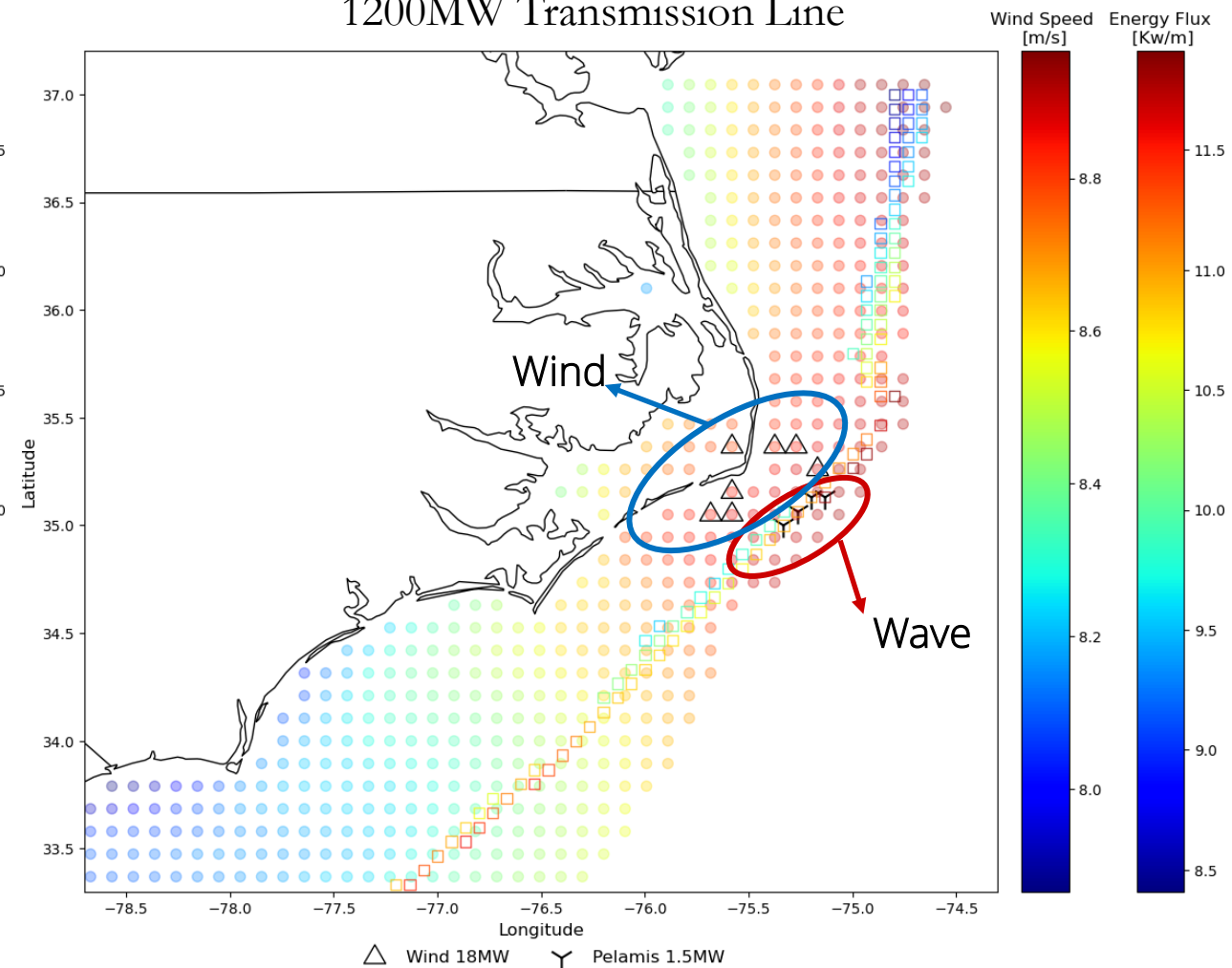


Where These Devices Are Deployed?

Optimistic-2050 Pelamis + 8MW at 115[\$/MWh] With
300MW Transmission Line



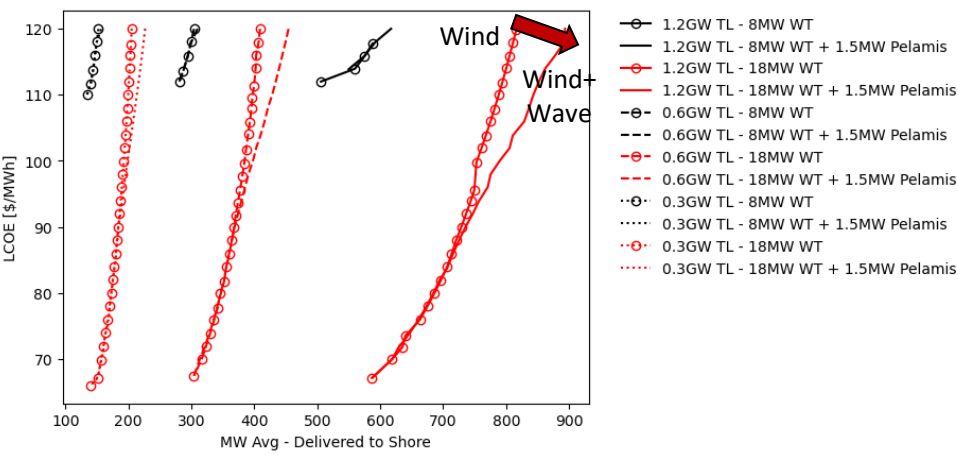
Optimistic-2050 Pelamis + 18MW at 80[\$/MWh] With
1200MW Transmission Line



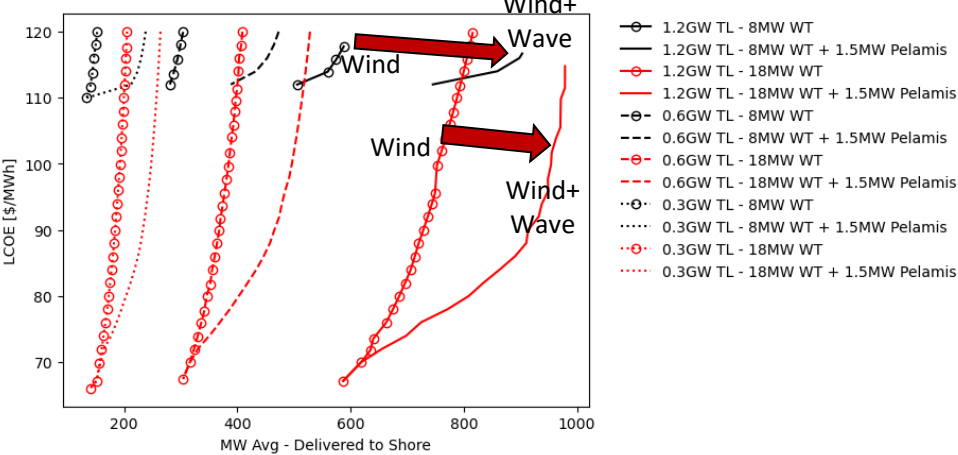
Portfolio Optimization Results

(BOEM LIMITED)

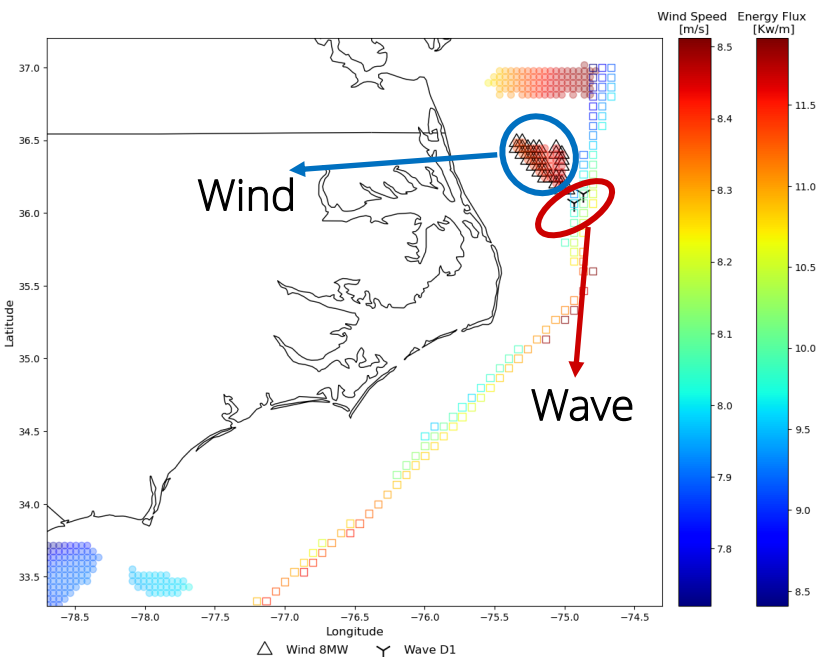
Conservative-2050 Pelamis
(2.5x reduction in cost)



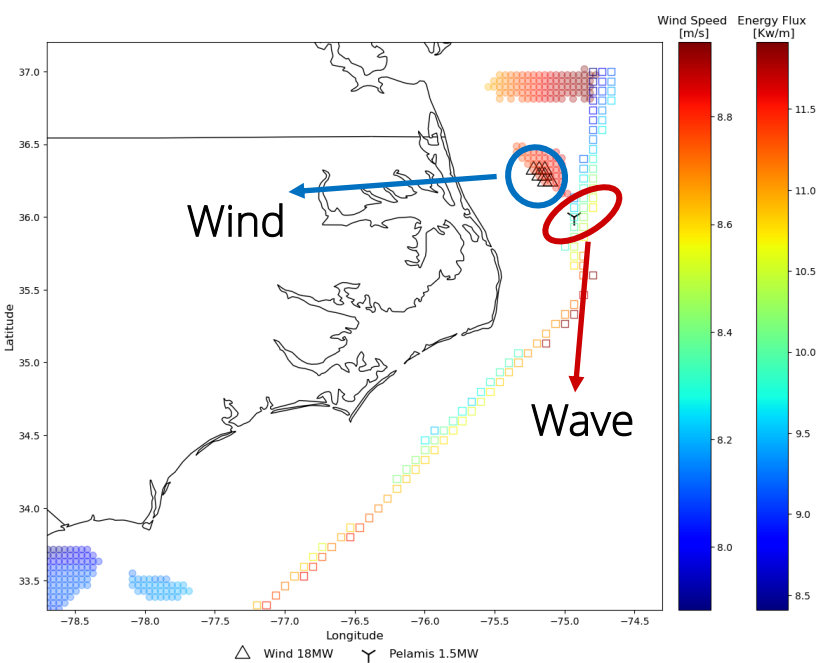
Optimistic-2050 Pelamis
(4.6x reduction in cost)



Optimistic-2050 Pelamis +
8MW Wind at
115[\$/MWh] With
300MW Transmission Line



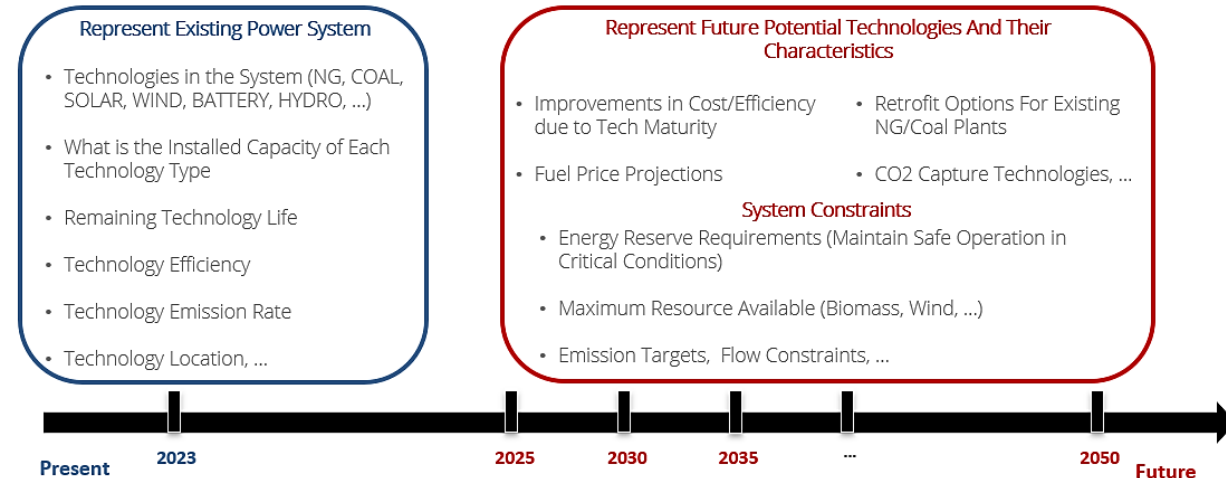
Optimistic-2050 Pelamis +
18MW Wind at
80[\$/MWh] With 300MW
Transmission Line



Capacity Expansion Model

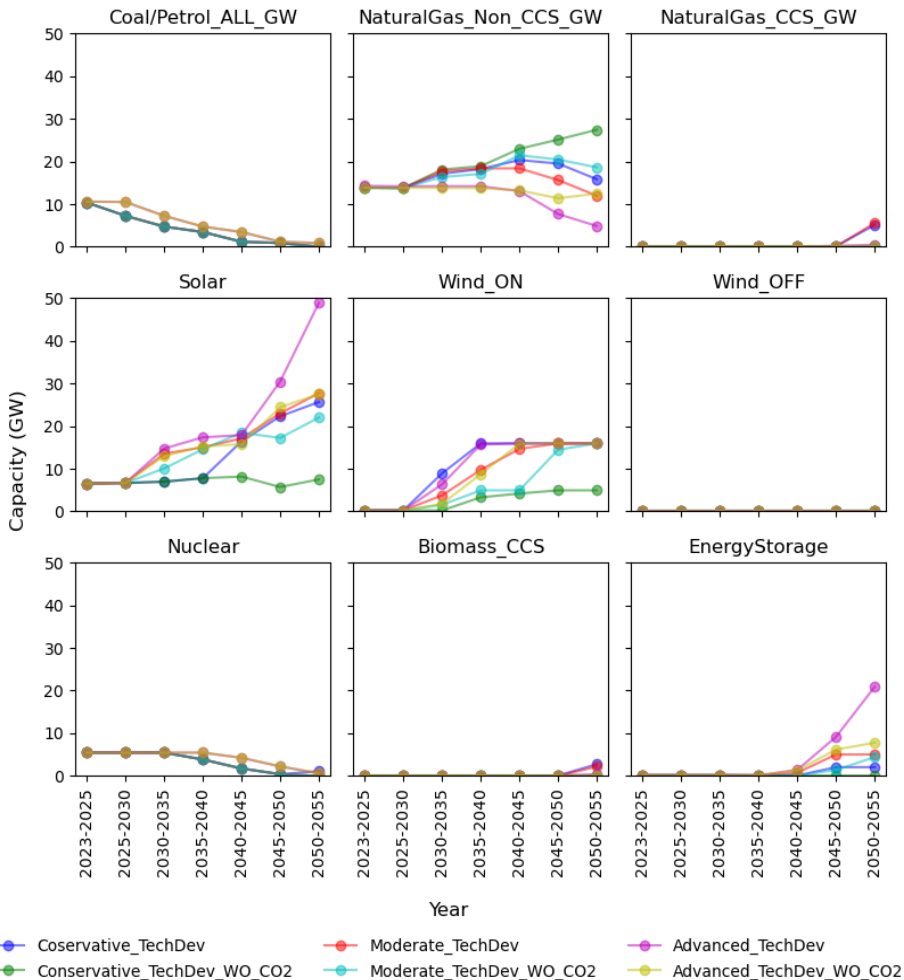
Minimize: TotalCost = LoanCost + FixedCost + VariableCost	Objective Function
$ACT_{p,s,d,t,v,r} = \sum_{(i,o)} FO_{p,s,d,t,v,r,i,o}$ $\forall \{p, s, d, t, v, r\} \in \Theta_{act}$	Process activity
$(CF_{s,d,t,v,r} \cdot C2A_t \cdot SEG_{s,d} \cdot TLF_{p,t,v,r}) \cdot CAP_{t,v,r} \geq ACT_{p,s,d,t,v,r} \quad \forall \{p, s, d, t, v, r\} \in \Theta_{act}$	Technology Capacity
$\sum_{(i,t,v)} FO_{p,s,d,t,v,r,i,c} = DEM_{p,c,r} \cdot DSD_{s,d,c,r}$ $\forall \{p, s, d, r, c \in C^d\} \in \Theta_{demand}$	Supply-demand
$FO_{p,s,d,t,v,r,i,o} = EFF_{t,v,i,o} \cdot FI_{p,s,d,t,v,r,i,o}$ $\forall \{p, s, d, t \in T - T^{st}, v, r, o\} \in \Theta_{flow}$	Process-Level Commodity Flow Constraint
$\sum_{(s,d,t,v,i,o)} (EAC_{t,v,r,i,o,e} \cdot FO_{p,s,d,t,v,r,i,o}) \leq ELM_{p,e,r}$ $\forall \{p, r, e \in C^e\} \in \Theta_{demand}$	Emission limit constraint
Other Constraints and Bounds: Capacity/Activity Limits, Storage, Life of Technology, Retrofit Options, Planning Reserve Margin (Capacity Credit), Curtailment, Energy Imp/Exp	

- Solution that Minimize the Cost of Meeting Future Energy Demand
- Optimize an energy system for a pre-determined future horizon to decide **when, how much, place, and type of investments to be made.**
- Consider physical characteristics, technical and the interactions among system elements, energy policy, and markets

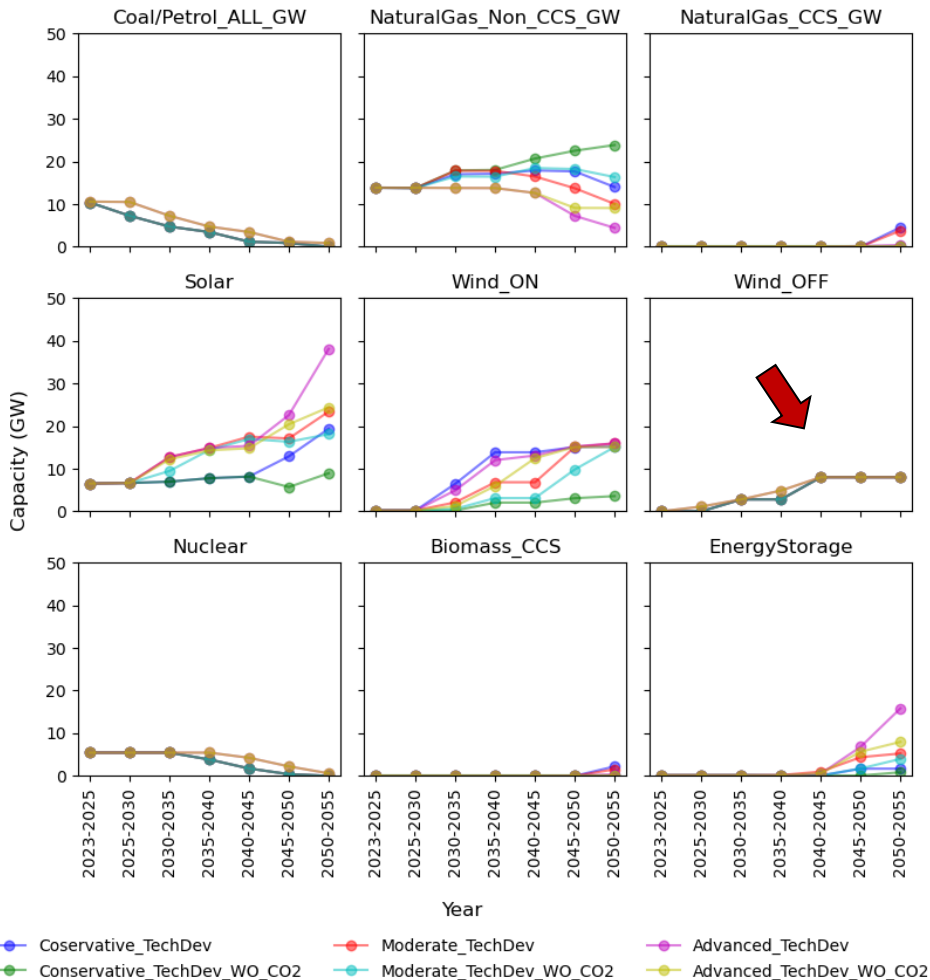


Capacity Expansion Model Results With and Without Offshore Wind Targets

System Capacity Without Offshore Wind Targets



System Capacity With Offshore Wind Targets



- Simulations With Conservative/Moderate/Advanced Tech Development from NREL-ATB; Simulations With and Without CO2-Targets/Offshore-Targets
- If we simulate the NC energy system without enforcing offshore wind targets (2.8GW – 2030, 8GW – 2040), the technology does not get deployed in the system, independently of the cost trajectory or CO2 targets.
- Enforcing an offshore wind target increases our costs by up to 7.7% from the reference case, which is reflected in solar and NG capacity reductions.

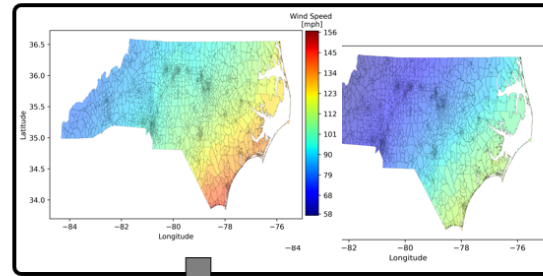
Percentage Difference In Obj Cost From Reference Case (Conservative, With CO2 Target, WO Wind Target)

Tech Trajectory	W/WO Wind Target	W/CO2 Target	WO/CO2 Target
Conservative	WO/Wind Target*	0.0%*	-4.6%
	W/Wind Target	7.7%	4.5%
Moderate	WO/Wind Target	-4.2%	-6.3%
	W/Wind Target	1.9%	0.4%
Advanced	WO/Wind Target	-8.3%	-10.2%
	W/Wind Target	0.8%	-0.4%

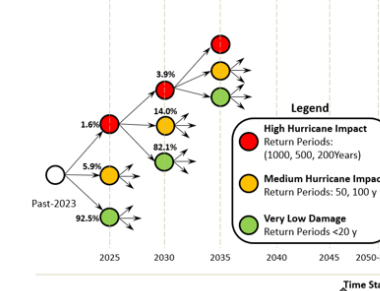
On Going Work (Q4)

- The Team will Perform Deterministic Capacity Expansion Simulations With Wind + Waves
- The Team Already Concluded Improving the Stochastic Capacity Expansion Model to Consider the Risk of Hurricane Damage to the Energy Infrastructure. This Model Will be Simulated in Q4 With Wave-Wind Energy Devices
- The Team is Working On The Construction of Fragility Curves for WEC Devices to be Used in the Stochastic Capacity Expansion Model

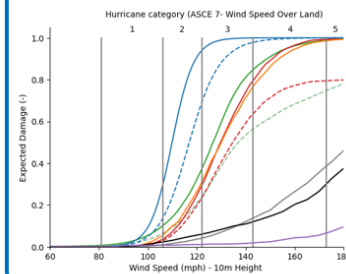
Modeling The Risk Of Hurricane Damage



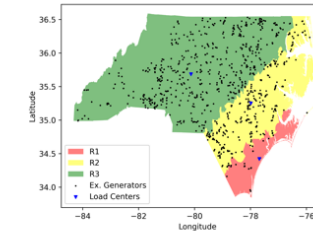
Scenario Tree



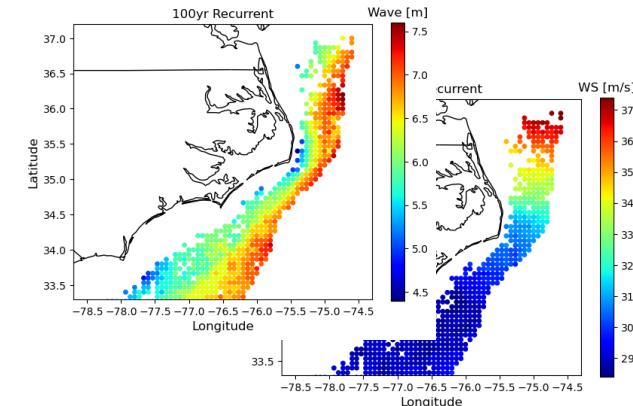
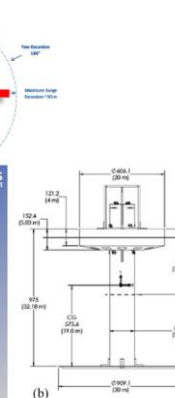
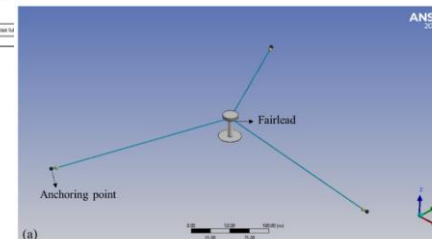
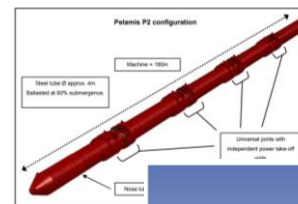
Technology Fragility Curves



**DAMAGE=Frangibility
(Wind Statistics)**



- Given different extreme wind scenarios with their corresponding probabilities, a stochastic optimization looks at each time step for the set of solutions (deployment size and type) that minimizes the **expected cost** to meet energy demand throughout the entire time horizon.
- In each scenario, damage to infrastructure is computed as the expected damage, conditioned on the hurricane intensities
- After the realization of a hurricane scenario, the existing capacity of any technology is reduced proportionally to its expected damage.





Thank you!

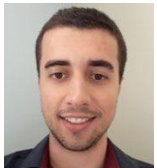
Any Questions?



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April, 2024